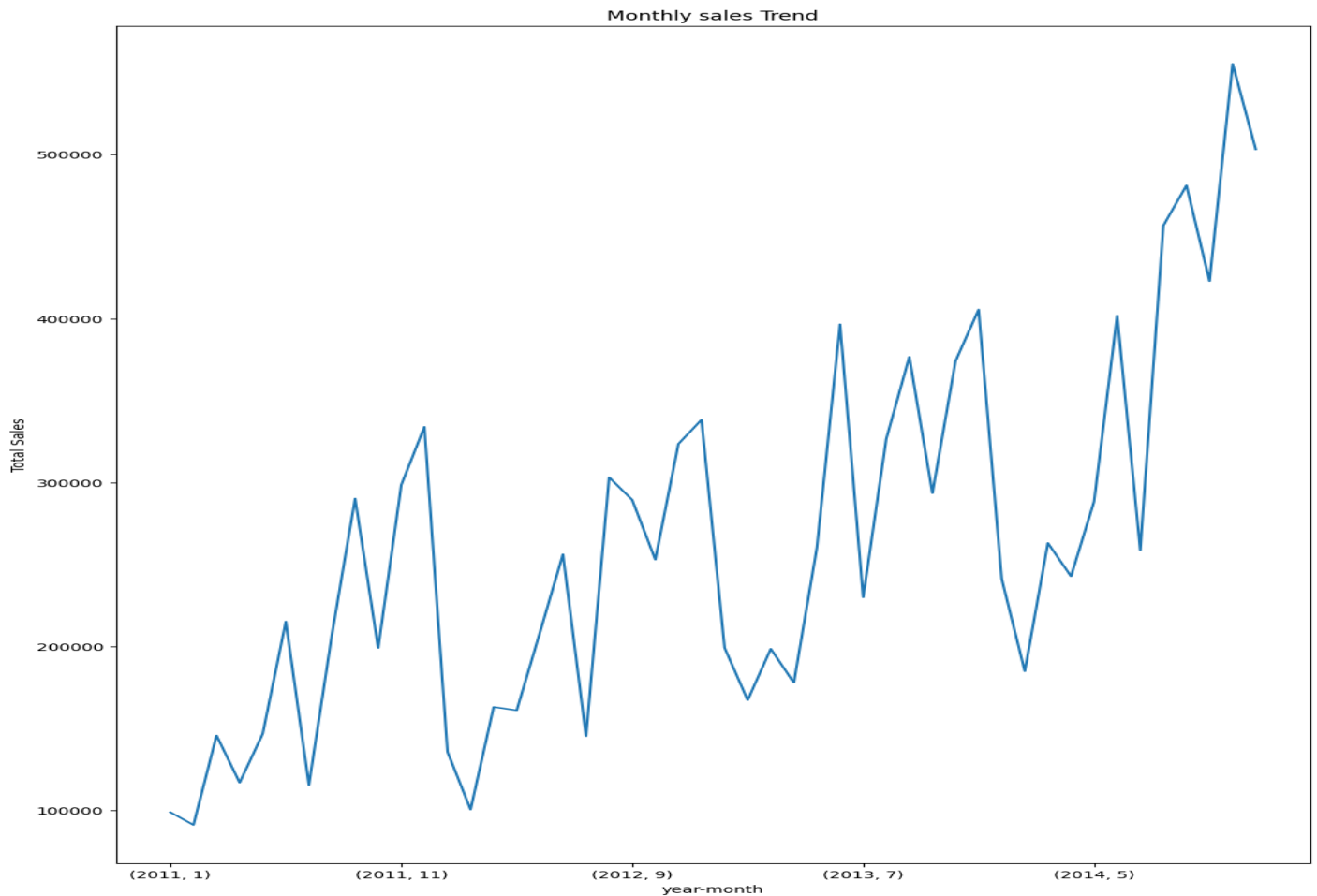


## E-Commerce Sales Analysis for Data-Driven Decision Making

### Week 2:

#### Sub task 1: Analyze Monthly and Yearly Sales Trends

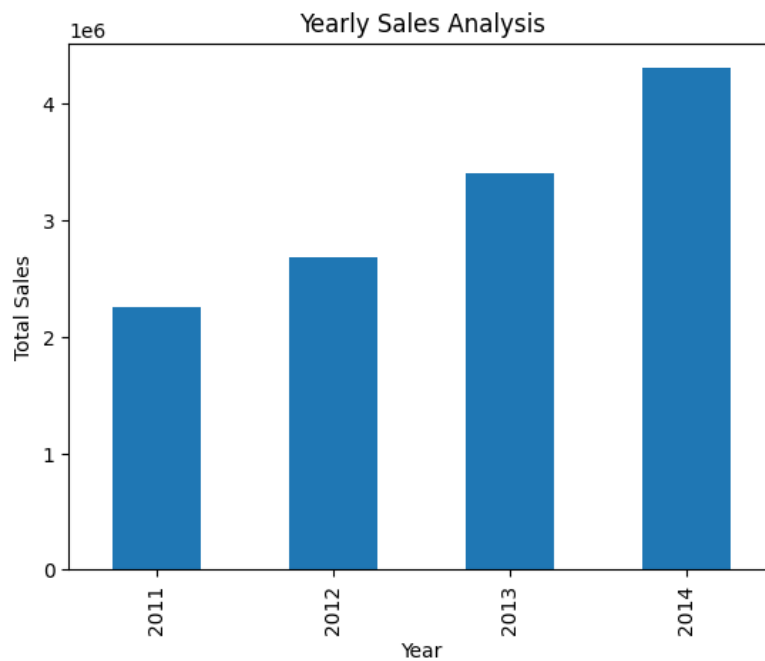


### Observation

#### Monthly Sales Trend Insights

- Sales showed a steady increase from **January 2011 to October 2011**.
- There was a noticeable drop in sales during **November 2011**.
- After the decline, sales started increasing again.
- Some fluctuations were observed across different months.
- The highest monthly sales were recorded in **2014**, showing strong business growth over time.

## YEARLY SALES



### OBSERVATION :

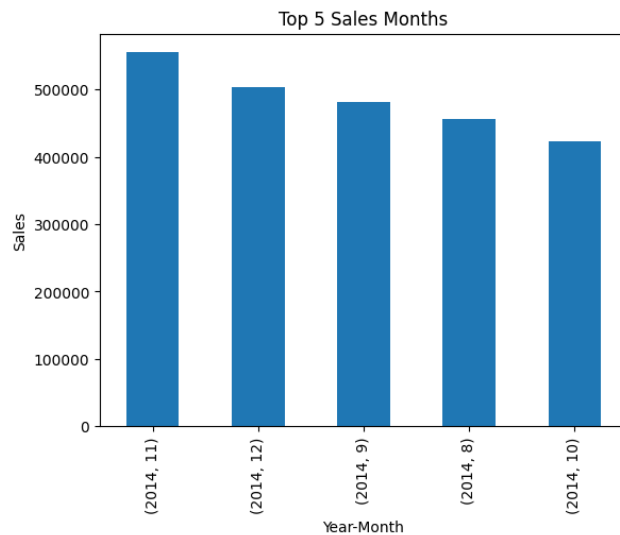
- 2014 recorded the highest sales.
- 2011 recorded the lowest sales.
- Sales showed consistent growth every year.
- No major yearly decline was observed.

This indicates the business was expanding steadily over time.

## Compare Peak and Low Sales Periods

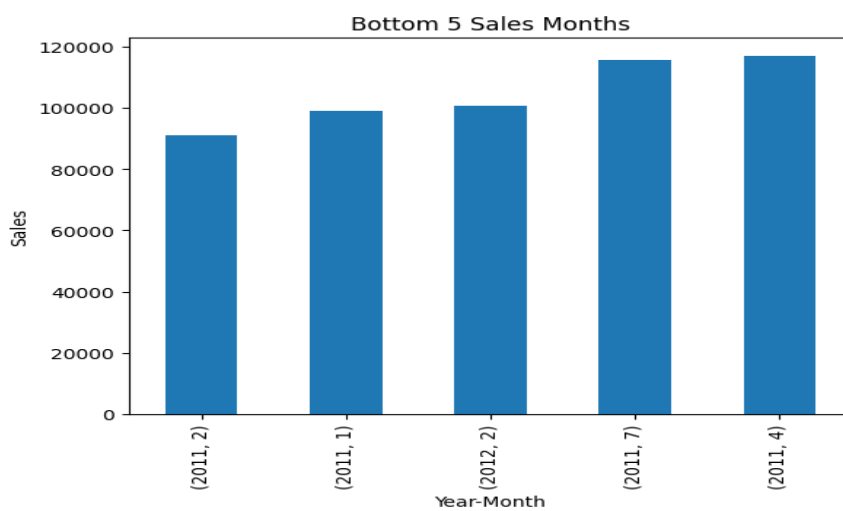
### Peak Sales Period

- Highest sales were recorded in **[2014,nov]**
- Possible reasons:
  - festive season sales
  - discounts/promotions
  - higher customer demand
  - holiday shopping



### Low Sales Period

- Lowest sales were recorded in **[2011,feb]**
- Possible reasons:
  - off-season demand
  - fewer promotions
  - lower customer spending



### Business Suggestions

- Offer discounts during low-performing months
- Increase marketing campaigns

- Launch seasonal promotions
- Improve customer retention offers

## **Sub task 2**

### **1. Product-Level Sales Analysis**

#### **Top Revenue-Generating Products:**

- Apple Smart Phone, Full Size – ₹86,936
- Cisco Smart Phone, Full Size – ₹76,441
- Motorola Smart Phone, Full Size – ₹73,159
- Nokia Smart Phone, Full Size – ₹71,904
- Canon imageCLASS 2200 Advanced Copier – ₹61,600

#### **Insights:**

- Smartphones contributed significantly to overall revenue.
- Technology products were highly demanded by customers.
- Expensive products generated higher revenue.

### **2. Low Performing Products**

Some products generated very low sales revenue.

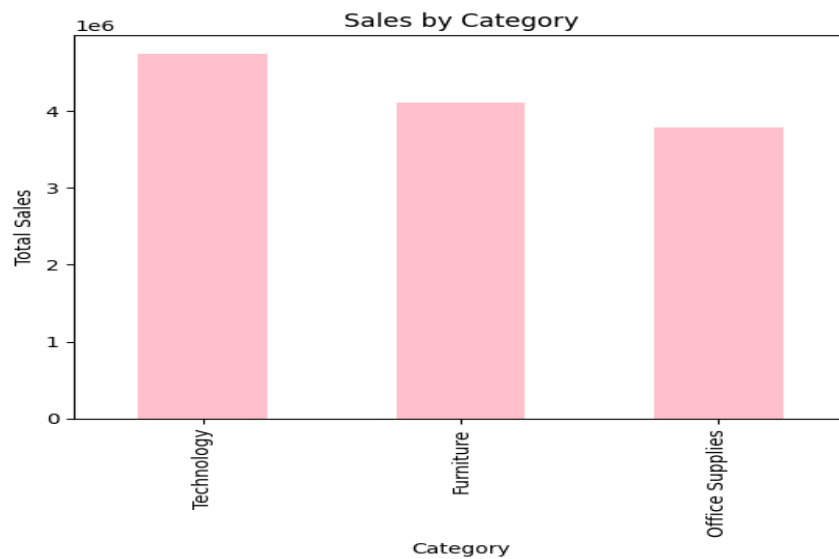
Examples:

- Avery 5 – ₹6
- Xerox 20 – ₹6
- Eureka Disposable Bags – ₹2

#### **Insights:**

- These products may have low customer demand.
- Poor product visibility or low necessity may be reasons.
- Businesses can reduce inventory for such products.

### 3. Category-Level Sales Analysis



#### Findings:

- Technology generated the highest revenue.
- Furniture generated moderate revenue.
- Office Supplies generated lower revenue compared to Technology.

#### Insights:

Technology products such as smartphones and electronics drive major sales because they have higher prices and strong customer demand.

### 4. Product Profitability Analysis

Sales alone do not determine business success. Profit analysis was conducted to identify highly profitable products.

The top-selling products were not always the most profitable products.

#### Insights:

This indicates:

- High discounts may reduce profit
- High shipping costs may reduce margins
- Some products sell more but earn less profit

## **Final Business Recommendations**

- Maintain sufficient stock for high-demand products.
- Reduce inventory for low-performing products.
- Improve pricing strategy for high-sales but low-profit products.
- Promote high-profit products through targeted marketing campaigns.
- Use demand forecasting to manage inventory efficiently.

## **Sub task 3**

### **Geographic Sales Performance Analysis**

#### **State-Level Analysis**

Sales were grouped by state to identify top-performing and low-performing states.

#### **Findings:**

- Top 10 states generated the highest revenue.
- These states represent strong customer demand.
- Bottom-performing states had significantly lower sales.

#### **Insight:**

High-performing states may have:

- larger customer base
- better logistics access
- higher online shopping adoption

Low-performing states may need:

- better marketing campaigns
- improved product availability

#### **City-Level Analysis**

City-wise sales analysis identified top and low-performing cities.

#### **Findings:**

- Major cities generated higher sales.
- Smaller cities contributed lower revenue.

### Insight:

Urban cities generally have higher purchasing power and better internet accessibility.

### Region-Level Analysis

Regions were compared using total sales revenue.

### Findings:

- Some regions contributed significantly more revenue.
- Certain regions showed weaker performance.

### Business Recommendations

- Focus marketing efforts in low-performing states
- Expand logistics network in underperforming regions
- Increase inventory in high-demand locations
- Use region-specific product promotions

